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Investigation of different effects on the capacity of supercapacitor comprising an ionic liquid between graphene oxide nanosheets

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Abstract

In this work, behavior of a supercapacitor has been simulated by confining the [EMIM][BF₄], [BMIM][BF₄], and [EMIM][PF₆] ionic liquids between two parallel graphene oxide (GO) surfaces. We have considered the different effects (such as the degree of oxidation, the pore size effect, and the effects of the kind of cation and anion) on the thermodynamics, dynamic, structural, and capacity of the supercapacitor. Our results showed that the interactions between the cations and anions are dominant at lower percent oxidation but, the interactions between the ions and the GO surfaces are dominant at higher percent oxidation. Also, there exists a competition between the columbic forces and the steric hindrances. The steric hindrances dominate at smaller percent oxidation but, the columbic forces dominate at higher degree of oxidation. The results showed that the self-diffusion increases by increasing the percent oxidation. The self-diffusion values also increase by increasing the pore size from 20 to 64 Å. The investigations indicated that the cell capacity decreases by increasing the oxidation degree of the electrodes. The cell capacity also increases by increasing the pore size. Also, the cell capacity using [BMIM][BF₄] is more than the [EMIM][BF₄] and the cell capacity using [EMIM][BF₄] is more than the [EMIM][PF₆].

1. Introduction

Storage of electric energy is important in many technologies such as electricity generation and electric vehicles.^{1,2} The electric double layer capacitors (supercapacitors), are significantly taken into consideration as energy storage devices in recent years.³⁻⁹ The supercapacitors have many benefits such as high-density cloud properties, high consistently stability, and wide temperature operation.^{5,10,11} Due to the high density power, the supercapacitors can be used in technologies where we need to attract or deliver high-load energy within a short time. By using the supercapacitors, many devices have also a long life cycle because only a small volume of the material is exposed to chemical decomposition of the electrolyte. The supercapacitors can be high-power generators and charge-discharge cycles without losing million dollars.¹²⁻¹⁴

A typical supercapacitor composes of two parallel electrodes and an electrolyte between the electrodes. According to the wide electrochemical window, low vapor pressure, and high thermal stability of ionic liquids (ILs), they are interesting alternatives to conventional electrolytes in the supercapacitors. Of course, the low ionic conductivity is the major disadvantage of the ILs.¹⁵

To improve the transport properties of charge carriers in supercapacitors, carbon electrodes in different forms have been developed such as carbon nanotubes and graphene, etc.⁵ Especially, the nano-layer graphene is suitable for use as electrodes because of its high electrical conductivity, low thermal and chemical resistance, and high capacity.^{15,16} Also, the flat structure of graphene can significantly facilitate ion transfer which increases the power density of a supercapacitor.¹⁵

The combination of ionic liquid electrolytes and graphene-based electrodes shows excellent performance of a supercapacitor. Recently, several experimental and simulation investigations have focused on these systems.^{3,17-25} However, the exactly characterizing chemical and spatial properties of the graphene-based electrodes is not easy.³ For this reason, many prior simulation studies have used the pure graphene as the electrodes in the supercapacitors.

Graphene oxide (GO) is a derivative of graphene but it have differing properties. Sheets of GO possess numerous oxygen containing functional groups: hydroxyl and carboxyl groups are located around the edges, whereas carbonyl and epoxide groups are in the center. The existence of various types of hydrophilic groups allows GO to be easily exfoliated state²⁶.

For the first time, Glukhova and et al. have considered the problem of chemical functionalization of mono- and bilayer graphene by epoxy and hydroxyl groups, taking into account a real geometry

profile of synthesized carbon objects²⁷. Also, a number of models have proposed for the GO structure, combining carboxyl, hydroxyl, carbonyl and epoxide groups at different locations by Chaban and Prezhdo²⁸.

Some investigations suggested that the graphene oxide (GO) electrodes have larger capacitance than the pure graphene electrodes.^{3,29,30} Also, reduced graphene oxide (RGO) which consists of a layer of graphene with makeup functional groups such as carbonyl, hydroxyl, epoxy, and ether groups has flexible applications and can be used as the carbon-based electrodes in the supercapacitors.^{3,31} Recently, DeYoung et al.³ simulated the supercapacitors with an ionic liquid ([EMI⁺][BF₄⁻]) and graphene oxide electrodes using molecular dynamics (MD) simulations. They just investigated the effect of electrode oxidation (using with hydroxyl groups) on capacitance of the supercapacitor.

2. Simulation method

In this work, behavior of the supercapacitor has been simulated by confining the [EMIM][BF₄], [BMIM][BF₄], and [EMIM][PF₆] ionic liquids between two parallel graphene oxide surfaces. We have considered the different effects (such as the degree of oxidation, the pore size effect, and the effects of the kinds of cation and anion) on the thermodynamics, dynamic, structural, and capacity of the supercapacitor.

We have used a one-layered GO with 450 carbon atoms, with OH groups of both sides. All of the MD simulations were carried out in the *NVT* ensemble with the Nosé-Hoover thermostat using the DL_POLY 4.03 software³² at 300 K. The OPLS-AA interaction potential has been used in this work which has been previously used in MD simulation of GO³³. In this work, the geometry optimization was performed using the DFT method by the B3LYP functional. This method was recently performed for GO³.

The model parameters for ionic liquids were derived from the force field recently developed by Roy and Maroncelli³⁴ and Merlet and et al.^{35, 36}. This interaction model can compute different properties of ionic liquids properly. The van der Waals and the electrostatic interactions have been considered in the system between the ionic liquid and the GO³. We have used the cutoff distance of the van der Waals interactions as 20 Å. The long-range electrostatics interactions have been calculated using the Ewald summation technique. The leapfrog-verlet algorithm has been used to

integrate the equations of motion with 1 fs of time step. The MD simulations were carried out for 5 ns of equilibration followed by production time of 40 ns for calculated properties.

Initial configuration of the system of parallel GOs was fixed at 20 and 64 Å and along the X-Y plane. Ionic liquid molecules filled the space between two surfaces of GO randomly in the area of L_z . Our simulation box was an orthorhombic cell with dimensions of $3.22 \times 3.22 \times 30.0 \text{ nm}^3$. The three dimensional periodic boundary conditions (pbc) were also applied³. The snapshots of the supercapacitor composed of graphene oxide with 30% oxidation and the two pore sizes of 20 and 64 Å have been presented in Fig.1.

The surface charge density, $\sigma = \pm 0.43 \text{ e nm}^{-2}$ corresponding to total electrode charges of $\pm 5e$, was considered by assigning partial charges to carbon atoms uniformly. The electrodes located at $Z = -Z_0$ and $+Z_0$ were charged positive (“anode”) and negative (“cathode”), respectively³.

3. Results and discussion

3.1 The degree of oxidation

In order to examine the effect of the degree of oxidation of the GO on the configurational energy of the system, we have confined the IL between the parallel GO surfaces with 20 Å distance and with the different amount of oxidation (0, 10, 30, 50, 70, 90, and 100 percent oxidation). The selection of these values of oxidation is according to the work of DeYoung et al.³. The configurational energy of the supercapacitor has been presented in Fig. 2. According to this figure, the absolute value of the configurational energy increases as the percent oxidation of GO increases. This increase can be due to the increasing of the interactions between the ions and the GO surfaces by increasing the percent oxidation. The snapshots of the IL confined between the GO surfaces with 70% and 90% oxidation have been presented in Fig. 3. According to this figure, by increasing the percent oxidation of the surfaces the interaction (and the number) of cations and anions in the middle of the confined space decreases but, the interaction between the ions and the GO surfaces increases.

To study the dynamical behavior of the cations and anions by changing the amount of oxidation on the GO surfaces, we have computed the self-diffusion coefficients of the confined ions. The self-diffusion coefficient has been calculated from the mean square displacement (MSD) of ions, using the Einstein equation:²⁹

$$D = \lim_{t \rightarrow \infty} \frac{1}{6tN} \sum_{i=1,N} \langle |r_i(t) - r_i(0)|^2 \rangle \quad (1)$$

where $r_i(0)$ and $r_i(t)$ are the particles positions at times $t=0$ and t , respectively. The self-diffusion coefficients of cations and anions at the different percent oxidation of the GO surfaces have been shown in Fig. 4. As this figure shows, the self-diffusion values of the ions initially decrease by 10% oxidation of the graphene sheets. This can be due to the increasing of interactions between the ions and the hydroxyl groups on the GO surfaces. Also, the hydroxyl groups which have been oriented into the confined space decrease the accessible volume for movement of the particles and therefore, the diffusion of ions decreases. It is also shown in Fig. 4 that the self-diffusion of the ions is then constant until the percent oxidation of 70. After that, the self-diffusion coefficients of anions and cations increase but they behave differently. If we increase the percent oxidation more (up to 70%) the interaction between the cations and anions decreases but the interaction between the ions and the GO surfaces increases (especially the interaction between the cations and the hydroxyl groups on the negative electrode). This lead to faster movement of the ions toward the charged surfaces and therefore, the self-diffusion increases. The interactions between the cations and anions are dominant at lower percent oxidation but, the interactions between the ions and the GO surfaces are dominant at higher percent oxidation.

To study the structural behavior of the system at different percent oxidation, we have computed radial distribution function (RDF or $g(r)$) of the confined cations and anions. The $g(r)$ can be defined as:²⁹

$$g(r) = \frac{1}{\rho^2} \left\langle \sum_i \sum_{j \neq i} \delta(r + r_i - r_j) \right\rangle \quad (2)$$

where the angle brackets denote a spherical average as well as the usual configurational average. The cation-anode (the negative electrode) and the anion-cathode (the positive electrode) RDFs have been shown at the different percent oxidation in Fig. 5. This figure shows that the first peak of the cation-anode RDF increases as the percent oxidation increases. This is due to increasing of the interaction between the cations and the hydroxyl groups on the anode surface by increasing the percent oxidation. It is also shown in Fig. 5 that the anion-cathode RDF peaks appear in the larger distances than those of the cation-anode RDF. This is due to the increasing of the repulsion between the anions and the hydroxyl groups by increasing the percent oxidation on the cathode surface. We

can also observe different behaviors for the first peak of the anion-cathode RDF relative to that of the cation-anode RDF by increasing the percent oxidation (some peaks increase and some decrease by increasing the percent oxidation). In practice, there is a competition between the anion-cation and the anion-cathode interactions. By increasing the percent oxidation, the hydroxyl groups increase on both electrodes and so, the negative charge increases on anode and the positive charge decreases on the cathode. Therefore, the cations tend to migrate more to the anode and subsequently the anions follow the cations and get far from the cathode. In this case, the interaction between the anions and the cathode decreases and the RDF decreases. This phenomenon occurs at smaller percent oxidation. At higher amount of oxidation, the cations move faster to anode and the anions lose the opportunity to follow them. Therefore, the only interaction which remains is the anion-cathode interaction. In this case, the interaction between the anions and the cathode surface increases and the RDF increases.

We have calculated the z-density values (the values of density perpendicular to the GO surface) for both cations and anions at the different percent oxidation and presented in Figs. 6 and 7, respectively. According to these figures, in pure graphene, the cations tend to migrate to the negative electrode (anode). Therefore, the density of cations increases beside the anode. The anions tend to the positive electrode and so, the density of anions increases beside the cathode. However, due to the strong columbic attraction between the cations and anions, some of the cations and anions remain together. This is why some smaller peaks are observed in the intermediate distances. At 10% oxidation, the peak of anions near cathode shows a small increase but, that of the cations near anode shows a small decrease relative to the pure graphene. This can be due to the fact that the cations of the IL are bigger than anions and therefore, there is more steric hindrance for cations when they approach to anode. Such as this situation exists for the 50% oxidation. But, we can observe a reverse trend at 90% oxidation. In the other words, the peak of anions near cathode decreases but, that of the cations near anode increases. This can be due to the increasing of negative charge on both cathode and anode (due to increasing the hydroxyl groups by increasing the oxidation). Increasing the negative charge on anode leads to attracting more cations but increasing the negative charge on cathode attracts less anions. This situation escalates at 100% oxidation. In fact, we have a competition between the columbic forces and the steric hindrances. The steric hindrance dominates at smaller percent oxidation but, the columbic forces dominate at higher degree of oxidation.

The voltage of the cell ($\Delta\Phi_{\text{cell}}$), which is the difference of the electric potential between the cathode and anode electrodes, and the change ($\Delta\Delta\Phi_{\text{cell}}$) can be calculated from the following equations ³:

$$\Delta\Phi_{\text{cell}} = \Phi(z = -z_0) - \Phi(z = z_0) \quad (3)$$

$$\Delta\Delta\Phi_{\text{cell}} = \Delta\Phi_{\text{cell}}(\text{charged}) - \Delta\Phi_{\text{cell}}(\text{discharged}) \quad (4)$$

The total capacitance of the supercapacitor (c_s^{tot}) is typically calculated from the following formula:

$$c_s^{\text{tot}} = \frac{|\sigma_s|}{\Delta\Delta\Phi_{\text{cell}}} \quad (5)$$

where σ_s is the surface charge density. We have calculated cell capacity at the different percent oxidation and presented in Table 1. According to this table, the cell capacity decreases by increasing the oxidation degree of the electrodes. This result can be understood according to Eq. 5. Also, the "overscreening" phenomenon is the reason of the decreasing of capacity by oxidation degree. The overscreening is observed when several electric layers are formed near the electrode surface. The increasing of the migration of ions toward the electrode surfaces by increasing the oxidation degree leads to this phenomenon³.

3.2 The pore size effect

To study the pore size effect (the distance between the GO sheets) on the configurational energy of the system, we have confined the IL between the parallel GO surfaces with 50% oxidation at two different distances of 20 and 64 Å. The configurational energy of the supercapacitor for the size of 20 Å was -40441 (kJ/mol) and for the size of 64 Å was -84270 (kJ/mol). According to these values, by increasing the size of the pore, the number of ions increases and therefore, the number of the different interactions in the system increases. This result can be better observed in Fig 8 in which we have presented the snapshots of the IL confined between the GO surfaces with 50% oxidation at two different distances of 20 and 64 Å.

The self-diffusion coefficients of the confined cations and anions have been calculated for both pore sizes and presented in Table 2. According to this table, the self-diffusion values increase by increasing the pore size from 20 to 64 Å. This can be due to the decreasing of the ion-surface interactions by increasing the distance of ions from the surface (increasing the pore size).

Therefore, the ions feel weaker interactions at larger pore size and therefore, the self-diffusion coefficient increases. It is also observed that the anions have smaller values of self-diffusion coefficient than the cations at any fixed pore size. This result which is in good agreement with the previous investigations³⁷⁻⁴¹ and can be due to the fact that the anions have the negative charge distributed among the four fluorine atoms of the head-group, which can induce some friction associated to the translational motion of these species.

To study the structural behavior of the system at different pore sizes, the RDFs of the confined cations and anions with the electrode surfaces have been calculated at two pore sizes of 20 and 64 Å and presented in Fig. 9. This figure shows that the RDF peaks at the smaller pore size are stronger than those of the larger size. This result is in agreement with our diffusion result and can be due to the increasing of the ion-surface interactions at smaller pore size. In the other words, the ions feel stronger interaction with the surface at smaller pore size and therefore, the probability of finding the ions increases around the surface.

We have calculated the z-density values for both cations and anions at 50% oxidation at both pore sizes and presented in Fig. 10. According to these figures at pure graphene, there is not much difference between the heights of the z-density of the ions at both pore sizes. It is also shown that the peaks between the two electrodes increase as the pore size increases. This can be due to formation of more ionic layers between the two surfaces by increasing the distance between them.

The cell capacity at 50% oxidation have been calculated at both pore sizes and presented in Table 3. According to this table, the cell capacity increases by increasing the size of the pore. This is in good agreement with the results of self-diffusion and radial distribution function. Also, the "overscreening" phenomenon is the reason of the decreasing of capacity at the size of 20 Å.

3.3 The kind of cation

In order to examine the effect of the kind of cation on the configurational energy of the system, we have confined [BMIM][BF₄] and [EMIM][BF₄] ILs between the parallel GO surfaces with 30 and 70% oxidation with the pore size of 20 Å. The configurational energies of the supercapacitor have been presented in Table 4. According to these values, the configurational energy values for [EMIM][BF₄] are greater than [BMIM][BF₄] at both percent oxidation. This can be due to the similarity of the size of the cation and anion in [EMIM][BF₄] ionic liquid which results more

coulombic attraction between these ions. We have also presented the snapshots of [EMIM][BF₄] confined between the GO surfaces with 70% oxidation at two different distances of 20 Å in Fig. 11.

The self-diffusion coefficients of the cations and anions from both ILs have been calculated for both percent oxidation and presented in Table 5. According to this table, the self-diffusion values of [EMIM][BF₄] are smaller than those of [BMIM][BF₄]. This result is in agreement with the energy results and can be due to the greater interactions between EMIM⁺ and BF₄⁻. This can be due to the similarity of the size of the cation and anion in [EMIM][BF₄] ionic liquid. But, BMIM⁺ is larger than EMIM⁺ and therefore, the dissimilarity of the size of the cation and anion in [EMIM][BF₄] decreases the interaction between the cation and anion and so, the diffusion coefficient increases.

In order to investigate the kind of cation on the structural properties of the systems, the RDFs of the different ions with the electrode surfaces have been calculated at two percent oxidation and presented in Fig. 12. According to this figure, the RDFs of [BMIM][BF₄] are greater than [EMIM][BF₄]. This result is in agreement with our diffusion and energy results. In the other words, the BMIM⁺ and BF₄⁻ ions have weaker interactions with each other (due to the size dissimilarity) and therefore, they have stronger interactions with the electrode surfaces and the ion-surface RDF increases.

We have calculated the z-density values for the different anions and cations at 30 and 70% oxidation and presented in Figs. 13 and 14, respectively. According to these figures, the peaks of cations and anions of [EMIM][BF₄] ionic liquid is greater than [BMIM][BF₄] near the electrode surfaces. This can be due to the fact that the EMIM⁺ has smaller size than BMIM⁺ and therefore, the EMIM⁺ can approach easier to the anode surface and leave the anion. Also, the left anion approaches easier to the cathode surface. It should be noted that the GO surface has several hydroxyl groups and so, the steric hindrance exists and the size of ion is important.

The cell capacity at 30 and 70% oxidation have been calculated for both ILs and presented in Table 6. According to this table, the cell capacity using [BMIM][BF₄] is more than the [EMIM][BF₄]. This result is in agreement with the results of self-diffusion and radial distribution function and can be due to the more separation between the cation and anion for [BMIM][BF₄] than [EMIM][BF₄].

3.4 The kind of anion

In order to examine the effect of the kind of anion on the configurational energy of the system, we have confined [EMIM][BF₄] and [EMIM][PF₆] ILs between the parallel GO surfaces with 30 and 90% oxidation with the pore size of 20 Å. The configurational energies of the supercapacitor have been presented in Table 7. According to these values, the configurational energy values for [EMIM][BF₄] are greater than [EMIM][PF₆] at both percent oxidation. This can be due to the similarity of the size of the cation and anion in [EMIM][BF₄] ionic liquid which results more columbic attraction between these ions. Also, PF₆⁻ has weaker interaction with the electrode surface than BF₄⁻ which is due to the greater steric hindrance for PF₆⁻ than BF₄⁻ (PF₆⁻ is bigger than BF₄⁻). We have also presented the snapshots of [EMIM][PF₆] confined between the GO surfaces with 30% oxidation at two different distances of 20 Å in Fig. 15.

The self-diffusion coefficients of the cations and anions from both ILs have been calculated for both percent oxidation and presented in Table 8. According to this table, the self-diffusion values of [EMIM][BF₄] are smaller than those of [EMIM][PF₆]. This result is in agreement with the energy results and can be due to the greater interactions between EMIM⁺ and BF₄⁻. This can be due to the similarity of the size of the cation and anion in [EMIM][BF₄] ionic liquid. But, PF₆⁻ is larger than BF₄⁻ and therefore, the dissimilarity of the size of the cation and anion in [EMIM][PF₆] decreases the interaction between the cation and anion and so, the diffusion coefficient increases.

In order to investigate the kind of anion on the structural properties of the systems, the RDFs of the different ions with the electrode surfaces have been calculated at two percent oxidation and presented in Fig. 16. This figure shows that the RDF peaks of the cations are almost the same. It is also shown that the RDF peaks of PF₆⁻ appear at the larger distances than BF₄⁻ which is due to the larger size of PF₆⁻ than BF₄⁻. It is also shown that the first RDF peak of PF₆⁻ decreases by increasing the percent oxidation which can be due to the increasing of steric hindrance by the hydroxyl group on the electrode surface.

We have calculated the z-density values for the different anions and cations at 30 and 90% oxidation in Figs. 17 and 18, respectively. The z-density peak of PF₆⁻ near the electrode surface decreases by increasing the percent oxidation from 30% to 90% which can be due to the increasing of steric hindrance by the hydroxyl group on the electrode surface. This situation also exists for BF₄⁻ but the

effect is smaller than PF_6^- . The same situation also exists for the cations from both ILs. In the other words, the steric hindrance increases by increasing the hydroxyl groups on the anode surface and so, the z-density of cations near the surface decreases.

The cell capacity at 30 and 90% oxidation have been calculated for both ILs and presented in Table 9. According to this table, the cell capacity using $[\text{EMIM}][\text{BF}_4]$ is more than the $[\text{EMIM}][\text{PF}_6]$.

4. Conclusion

In this work, the supercapacitor has been simulated by confining the $[\text{EMIM}][\text{BF}_4]$ IL between the parallel GO surfaces. We have considered the different effects (such as the degree of oxidation, the pore size effect, and the effects of the kind of cation and anion) on the thermodynamics, dynamic, structural, and capacity of the supercapacitor.

The examination of the degree of oxidation showed that the absolute value of the configurational energy increases as the percent oxidation of GO increases. This increase can be due to the increasing of the interactions between the ions and the hydroxyl groups on the GO surfaces by increasing the percent oxidation. The hydroxyl groups also decrease the accessible volume for movement of the particles and therefore, the diffusion of ions decreases. If we increase the percent oxidation more (up to 70%) the interaction between the cations and anions decreases but the interaction between the ions and the GO surfaces increases and therefore, the self-diffusion increases. The interactions between the cations and anions are dominant at lower percent oxidation but, the interactions between the ions and the GO surfaces are dominant at higher percent oxidation. Also, there is a competition between the columbic forces and the steric hindrances. The steric hindrance dominates at smaller percent oxidation but, the columbic forces dominate at higher degree of oxidation. The results also showed that the cell capacity decreases by increasing the oxidation degree of the electrodes which is due to the "overscreening" phenomenon.

The investigation of the pore size showed that the number of ions increases by increasing the pore size and therefore, the number of the different interactions in the system increases. Also, the self-diffusion values increase by increasing the pore size from 20 to 64 Å. This can be due to the decreasing of the ion-surface interactions by increasing the distance of ions from the surface (increasing the pore size). The cell capacity also increases by increasing the size of the pore. This

is in good agreement with the results of self-diffusion and radial distribution function. Also, the "overscreening" phenomenon is the reason of the decreasing of capacity at the size of 20 Å.

The study of the cation effect showed that the configurational energy values for [EMIM][BF₄] are greater than [BMIM][BF₄] at different percent oxidations. This can be due to the similarity of the size of the cation and anion in [EMIM][BF₄] ionic liquid which results more columbic attraction between these ions. Also, the self-diffusion values of [EMIM][BF₄] are smaller than those of [BMIM][BF₄]. This result is in agreement with the energy results and can be due to the greater interactions between EMIM⁺ and BF₄⁻. The cell capacity using [BMIM][BF₄] is also more than the [EMIM][BF₄]. This result is in agreement with the results of self-diffusion and radial distribution function and can be due to the more separation between the cation and anion for [BMIM][BF₄] than [EMIM][BF₄].

The study of the anion effect showed that the configurational energy values for [EMIM][BF₄] are greater than [EMIM][PF₆] at different percent oxidations. This can be due to the similarity of the size of the cation and anion in [EMIM][BF₄] ionic liquid which results more columbic attraction between these ions. Also, PF₆⁻ has weaker interaction with the electrode surface than BF₄⁻ which is due to the greater steric hindrance for PF₆⁻ than BF₄⁻ (PF₆⁻ is bigger than BF₄⁻). Also, the self-diffusion values of [EMIM][BF₄] are smaller than those of [EMIM][PF₆]. This result is in agreement with the energy results and can be due to the greater interactions between EMIM⁺ and BF₄⁻. The cell capacity using [EMIM][BF₄] is also more than the [EMIM][PF₆].

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Table 1. The cell capacity of the supercapacitor at the different percent oxidation

% Oxidation	Capacity (Microfarad/Cm ²)
0	3.12
10	2.81
30	2.47
50	2.21
70	1.99
90	1.87
100	1.8

Table 2. The self-diffusion coefficients of the confined cations and anions at both pore sizes

Ions	D ($10^{-10}\text{cm}^2\text{s}^{-1}$)
BF_4^- (20 Å)	0.00011521
BF_4^- (64 Å)	0.00597540
BMIM^+ (20 Å)	0.00014853
BMIM^+ (64 Å)	0.00758440

Table 3. The cell capacity at both pore sizes.

BMIM BF₄	Capacity (MicroFarad/Cm ²)
Size 20	2.21
Size 64	2.43

Table 4. The configurational energies of the two different ILs into the supercapacitor

	Oxidation Percent	E (kJmol ⁻¹)
EMIMBF ₄	70	-53142.0
	30	-36275.0
BMIMBF ₄	70	-49778.0
	30	-32985.0

Table 5. The self-diffusion coefficients of the cations and anions from the different ILs

Oxidation Percent	Ions	D ($10^{-10}\text{cm}^2\text{s}^{-1}$)
70	BF_4^- (EMIM $^+$ BF $_4^-$)	0.00001632
	BF_4^- (BMIM $^+$ BF $_4^-$)	0.00011026
	EMIM $^+$	0.00012670
	BMIM $^+$	0.00011912
30	BF_4^- (EMIM $^+$ BF $_4^-$)	0.00008342
	BF_4^- (BMIM $^+$ BF $_4^-$)	0.00025084
	EMIM $^+$	0.00015522
	BMIM $^+$	0.00029259

Table 6. The cell capacity at 30 and 70% oxidation for both ILs

70% Oxidation	IL	Capacity (MicroFarad/cm ²)
	BMIM BF ₄	1.99
	EMIM BF ₄	1.95
30% Oxidation	BMIM BF ₄	2.47
	EMIM BF ₄	2.43

Table 7. The configurational energies of the two different ILs into the supercapacitor

	Oxidation Percent	E (kJmol ⁻¹)
[EMIM][BF ₄]	90	-63372.0
	30	-36275.0
[EMIM][PF ₆]	90	-58327.0
	30	-32925.0

Table 8. The self-diffusion coefficients of the cations and anions from the different ILs

Oxidation	Ions	D ($10^{-10}\text{cm}^2\text{s}^{-1}$)
90%	EMIM ⁺ (EMIM ⁺ BF ₄ ⁻)	0.000023812
	EMIM ⁺ (EMIM ⁺ PF ₆ ⁻)	0.000099668
	BF ₄ ⁻	0.000011222
	PF ₆ ⁻	0.0286000000
30%	EMIM ⁺ (EMIM ⁺ BF ₄ ⁻)	0.0001552176
	EMIM ⁺ (EMIM ⁺ PF ₆ ⁻)	0.0001433122
	BF ₄ ⁻	0.0000834242
	PF ₆ ⁻	0.0001142863

Table 9. The cell capacity at 30 and 90% oxidation for both ILs

30%	IL	Capacity (MicroFarad/cm ²)
	EMIM BF4	2.43
	EMIM PF6	2.21
90%	EMIM BF4	1.83
	EMIM PF6	1.69

Figure captions

- Fig. 1.** The snapshots of the supercapacitor composed of graphene oxide with 30% oxidation and the two pore sizes of 20 and 64 Å.
- Fig. 2.** The configurational energy of the supercapacitor with the different degree of oxidation (The IL is [BMIM][BF₄] and the pore size is 20 Å).
- Fig. 3.** The snapshots of the IL [BMIM][BF₄] confined between the GO surfaces with 70% and 90% oxidation with the pore size of 20 Å at before and after the simulation.
- Fig. 4.** The self-diffusion coefficients of cations and anions at the different percent oxidation of the GO surfaces (The IL is [BMIM][BF₄] and the pore size is 20 Å).
- Fig. 5.** The cation-anode and the anion-cathode RDFs at the different percent oxidation (The IL is [BMIM][BF₄] and the pore size is 20 Å).
- Fig. 6.** The z-density values for the anions at the different percent oxidation (The IL is [BMIM][BF₄] and the pore size is 20 Å).
- Fig. 7.** The z-density values for the anions at the different percent oxidation.
- Fig. 8.** The snapshots of the [BMIM][BF₄] confined between the GO surfaces with 50% oxidation at two different distances of 20 and 64 Å at before and after the simulation.
- Fig. 9.** The RDFs of the confined cations and anions of the IL [BMIM][BF₄] with the electrode surfaces with 50% oxidation at two pore sizes of 20 and 64 Å.
- Fig. 10.** The z-density values for both cations and anions of the supercapacitor composed of the IL [BMIM][BF₄] and the electrodes of 50% oxidation at two pore sizes of 20 and 64 Å.
- Fig. 11.** The snapshots of [EMIM][BF₄] IL confined between the GO surfaces with 70% oxidation at two distances of 20 Å.
- Fig. 12.** The RDFs of the different ions with the electrode surfaces have been calculated at two percent oxidation.
- Fig. 13.** The z-density values for the different anions at 30 and 70% oxidation.
- Fig. 14.** The z-density values for the different cations at 30 and 70% oxidation.
- Fig. 15.** The snapshots of [EMIM][PF₆] confined between the GO surfaces with 30% oxidation at two different distances of 20 Å.
- Fig. 16.** The RDFs of the different ions with the electrode surfaces at two percent oxidation.

Fig. 17. The z-density values for the different anions at 30 and 90% oxidation.

Fig. 18. The z-density values for the different cations at 30 and 90% oxidation.

ACCEPTED MANUSCRIPT

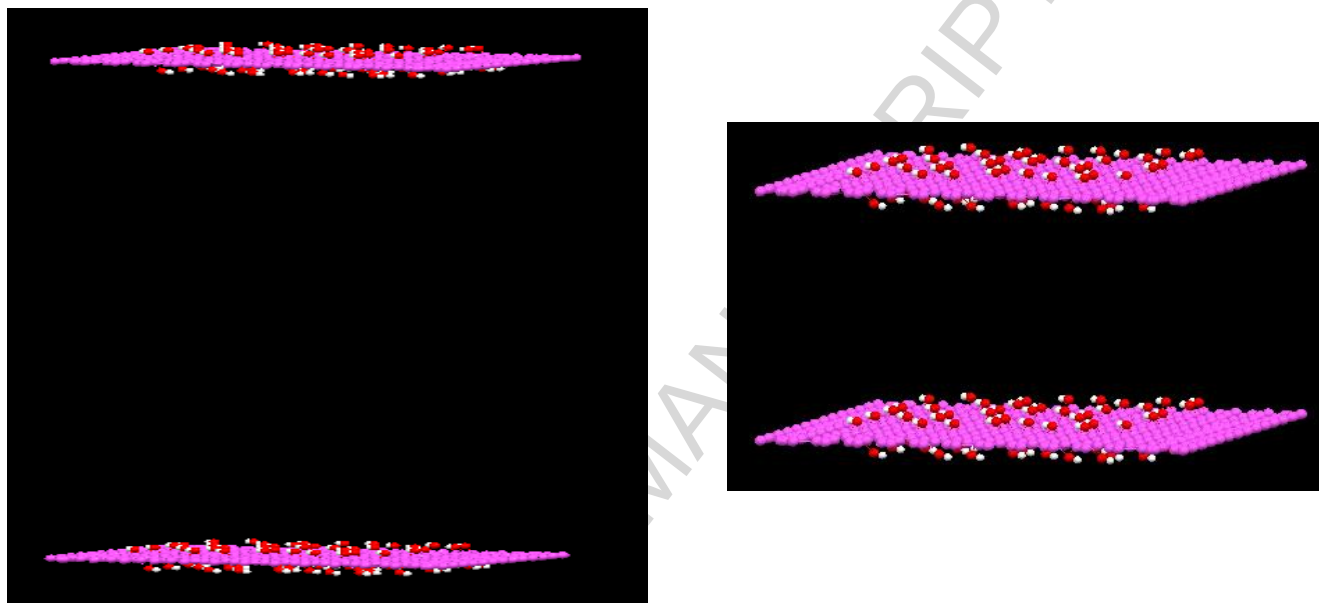
Fig. 1

Fig. 2

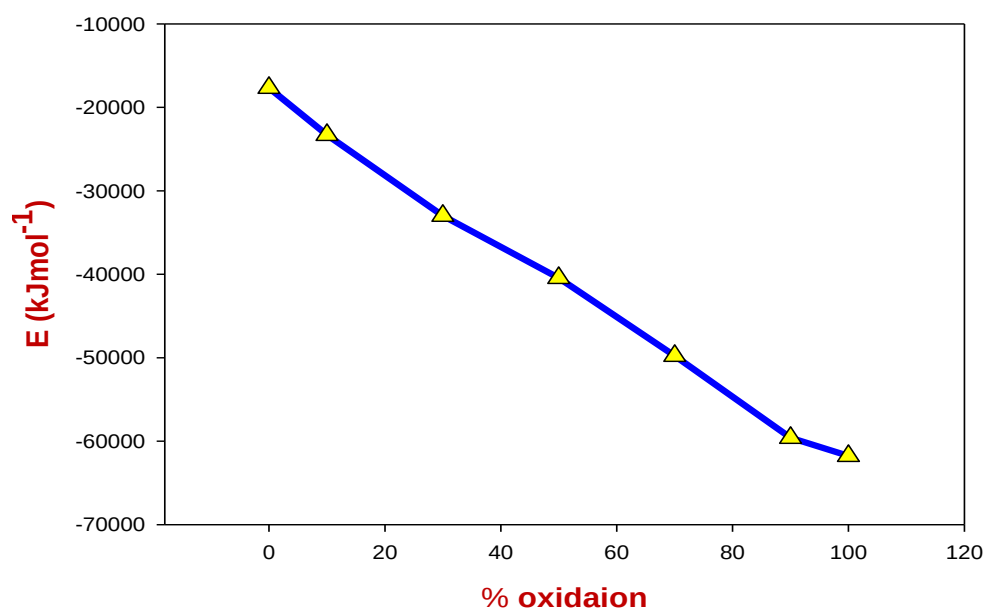


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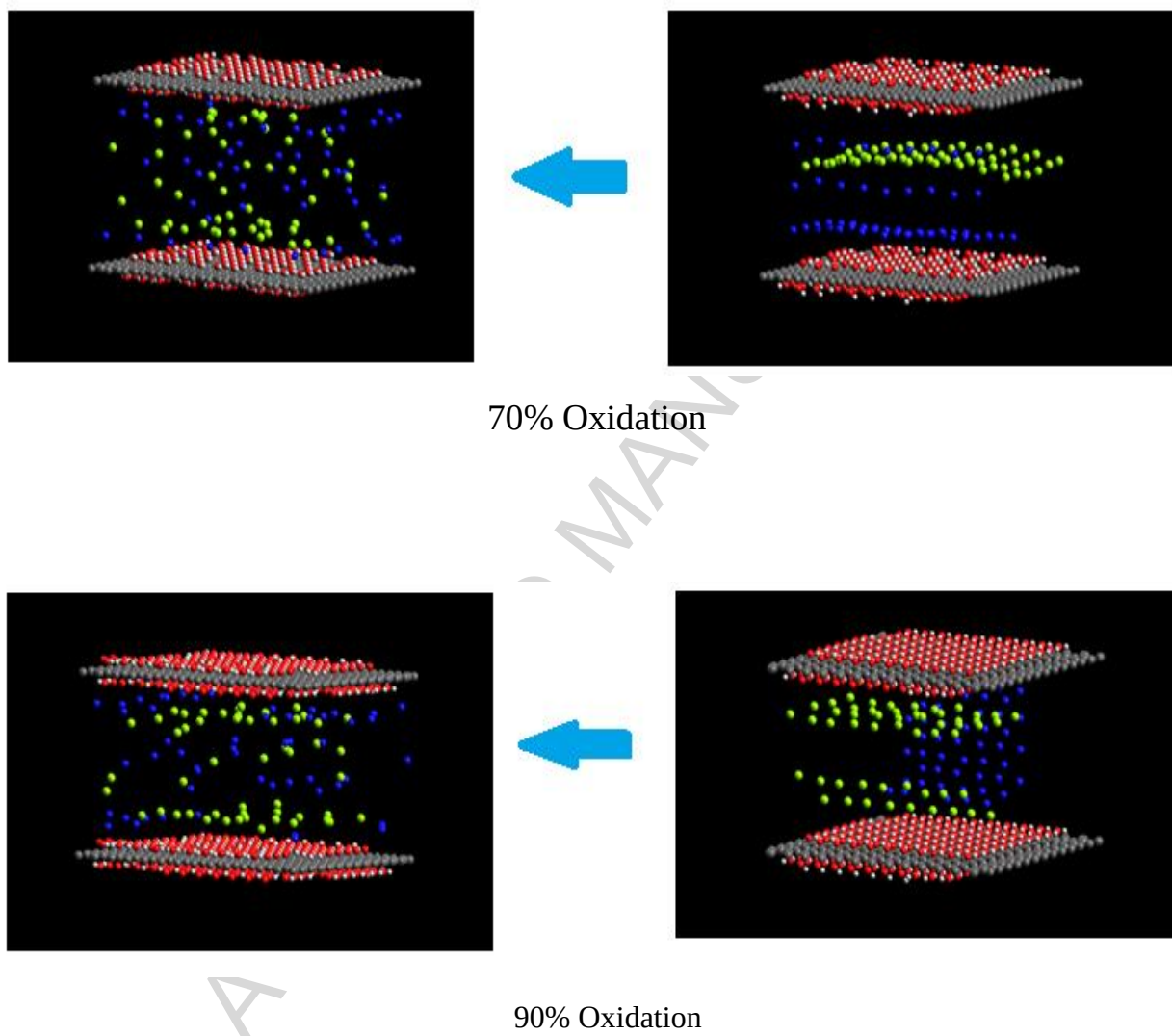


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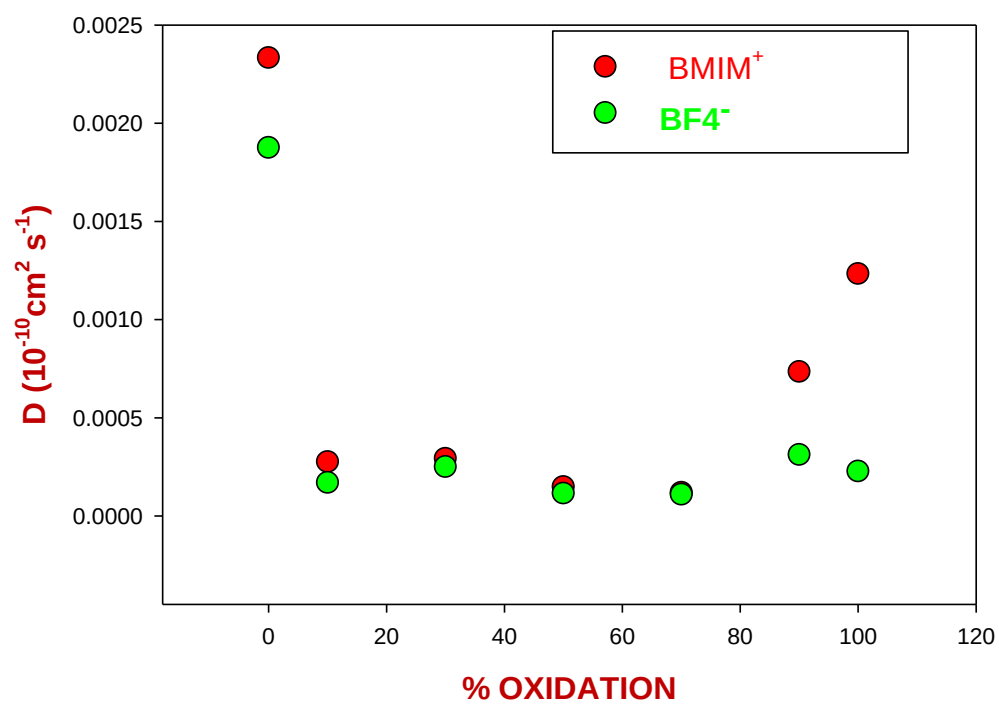


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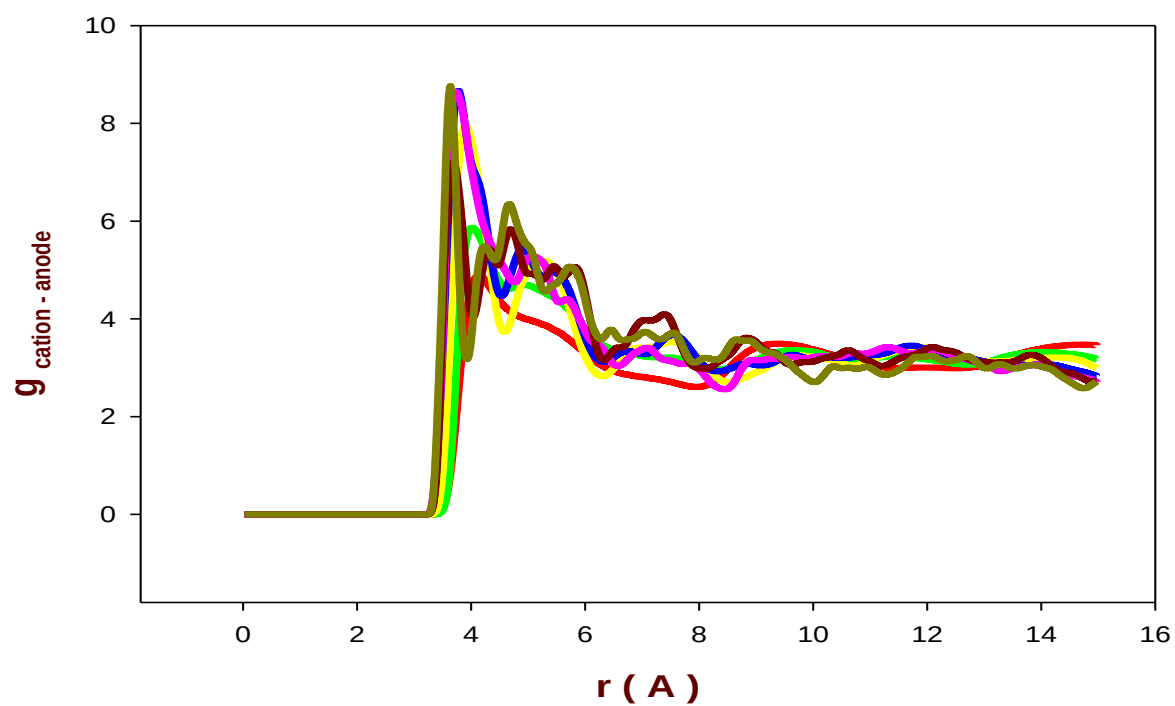
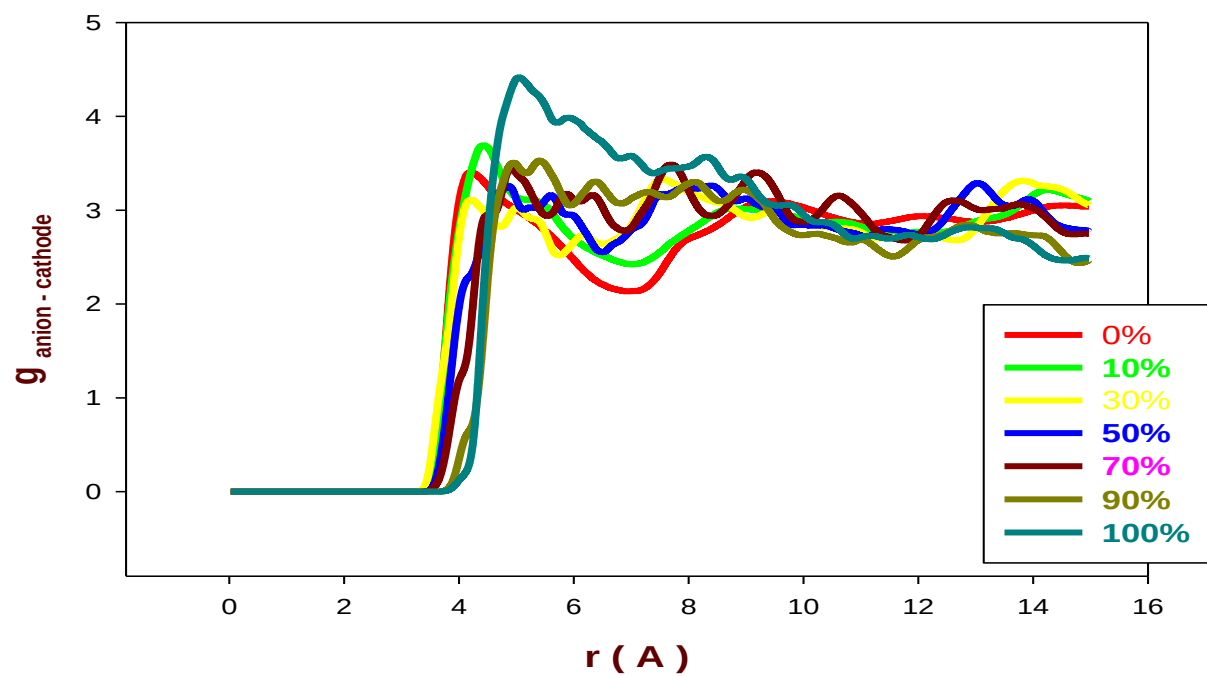


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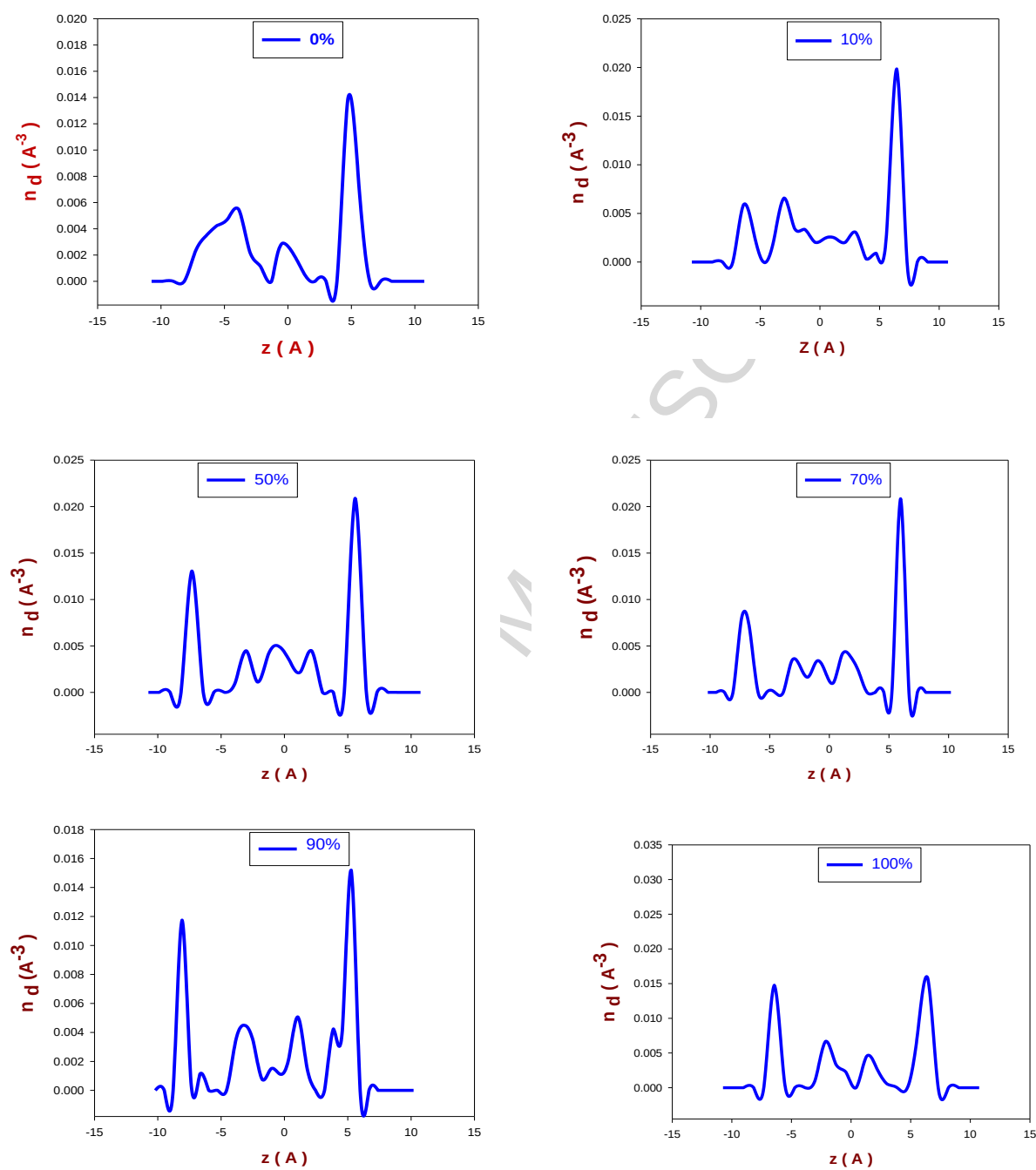


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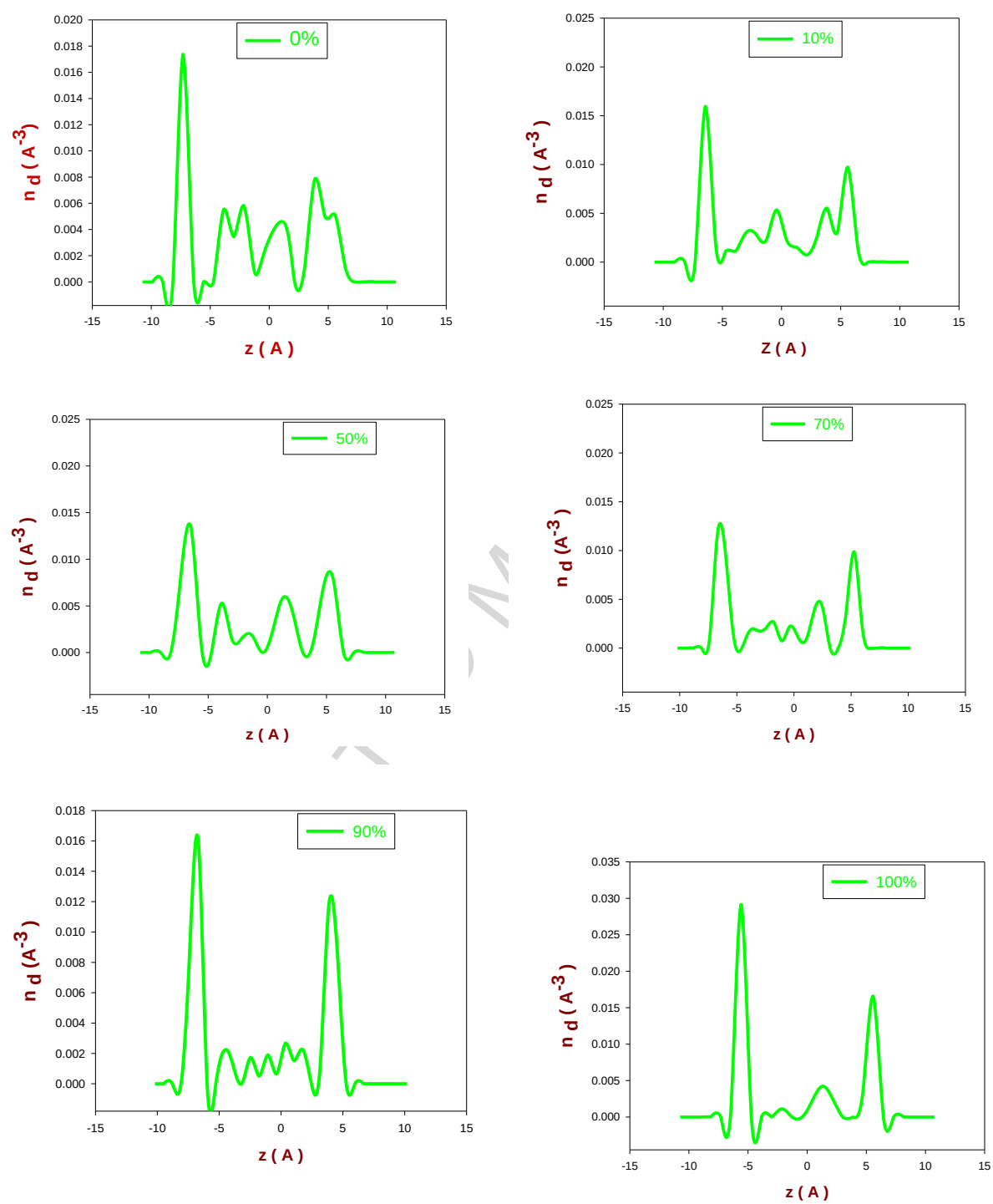


Fig. 8

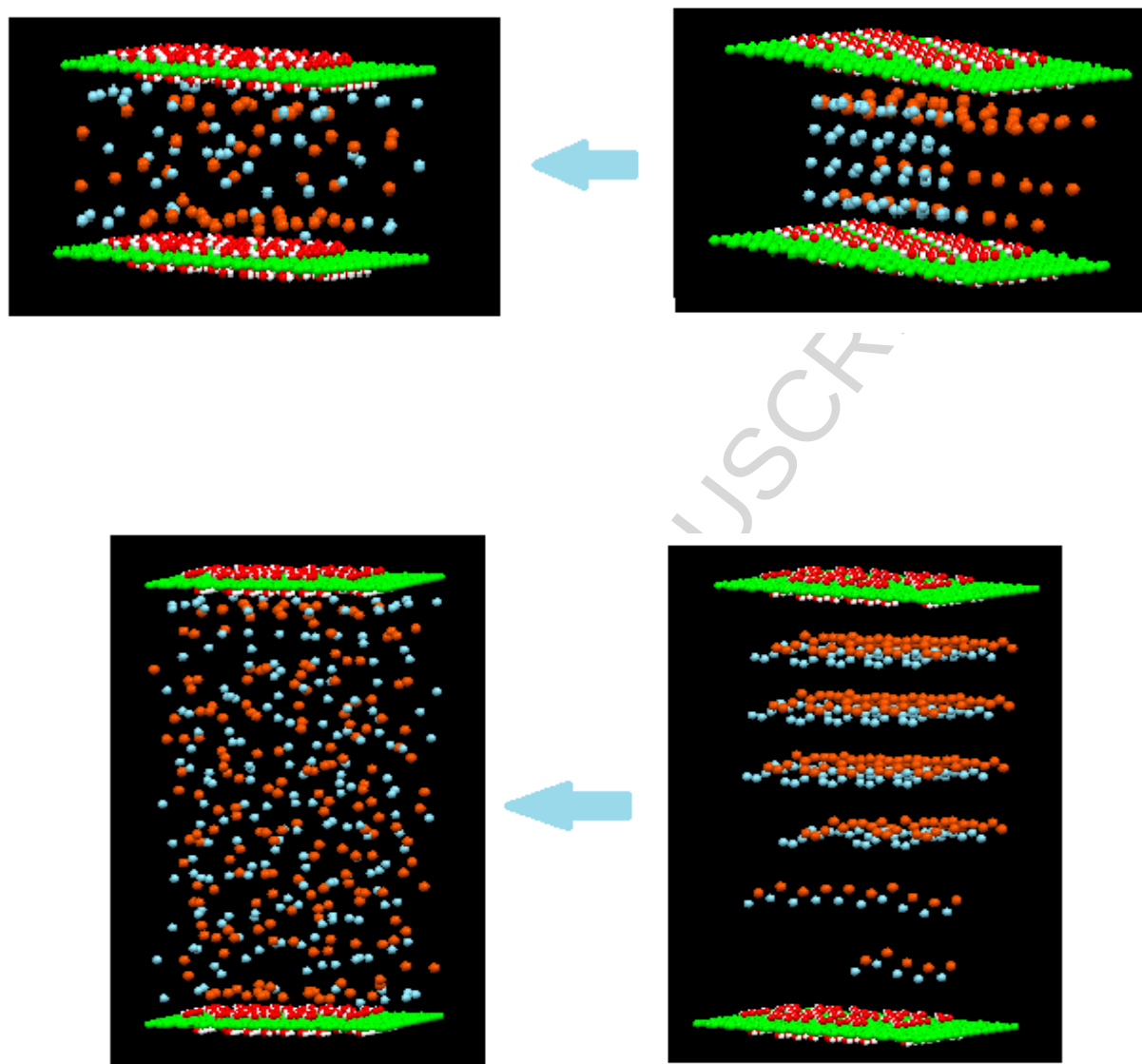


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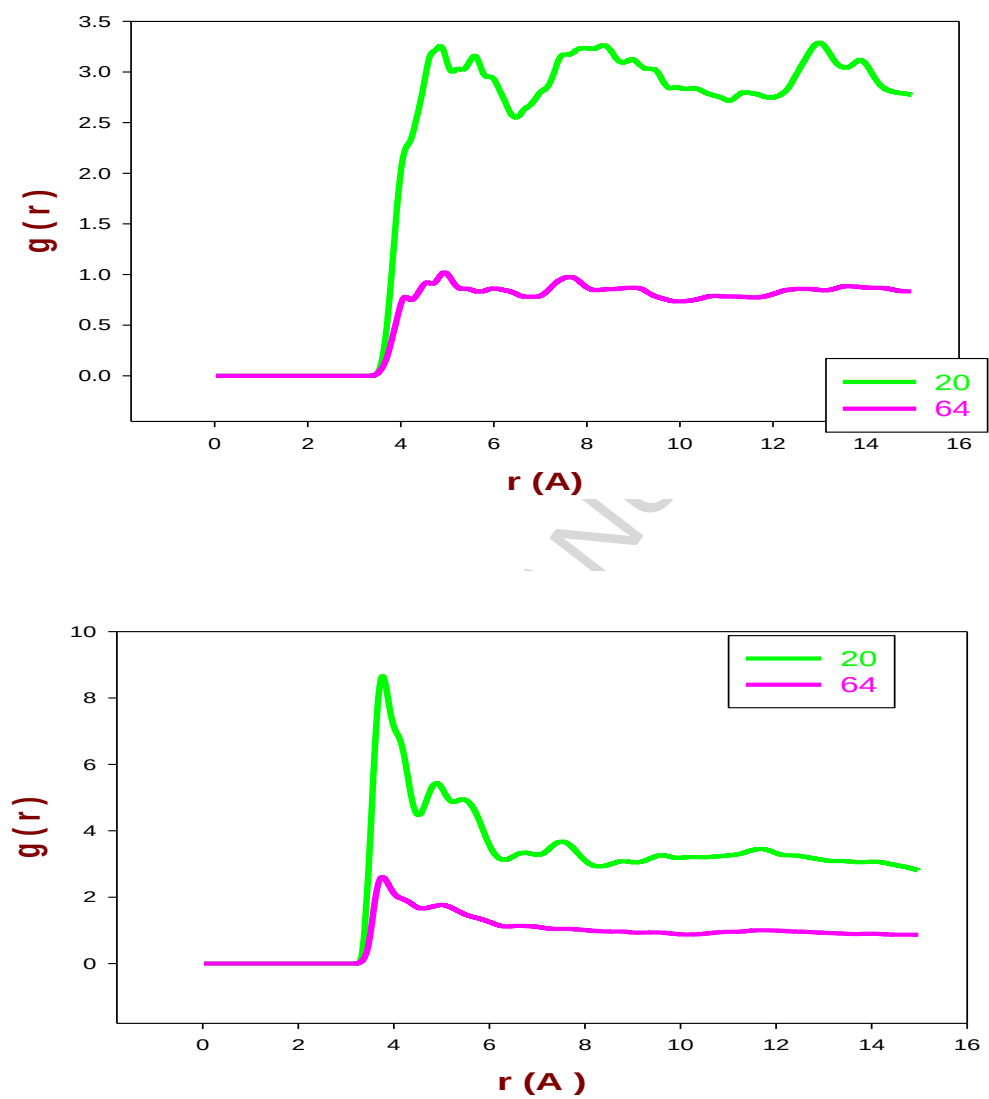


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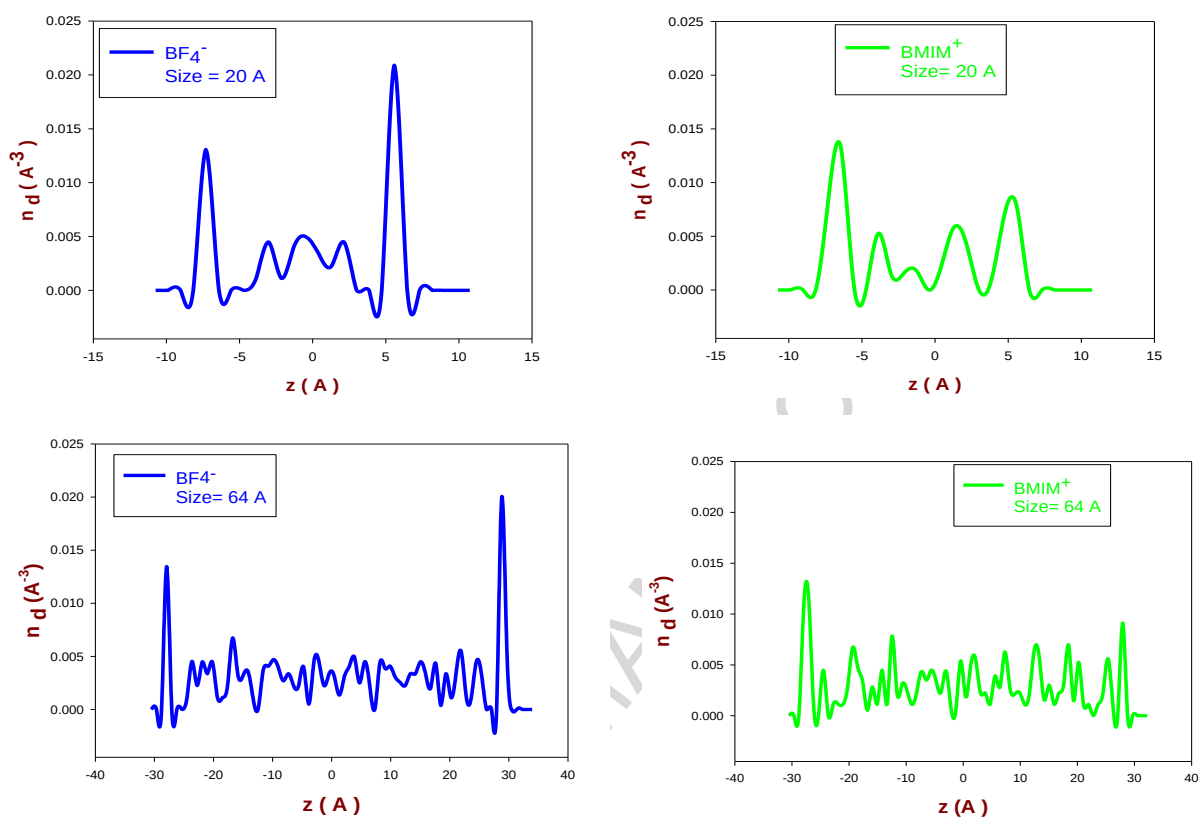


Fig. 11

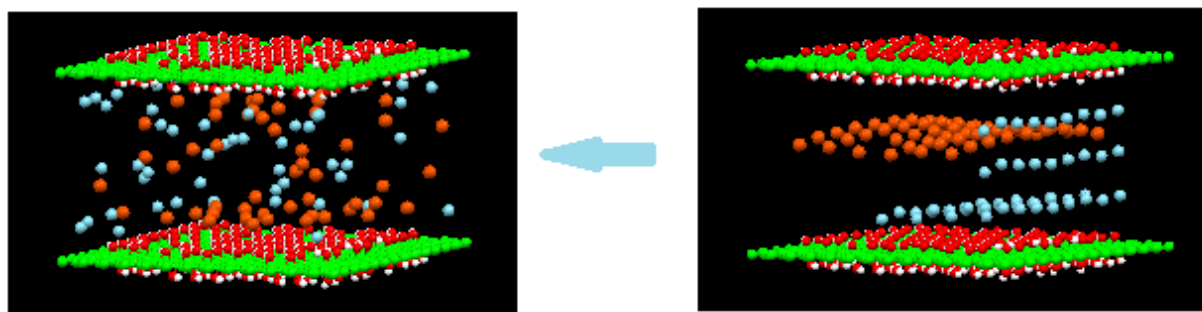


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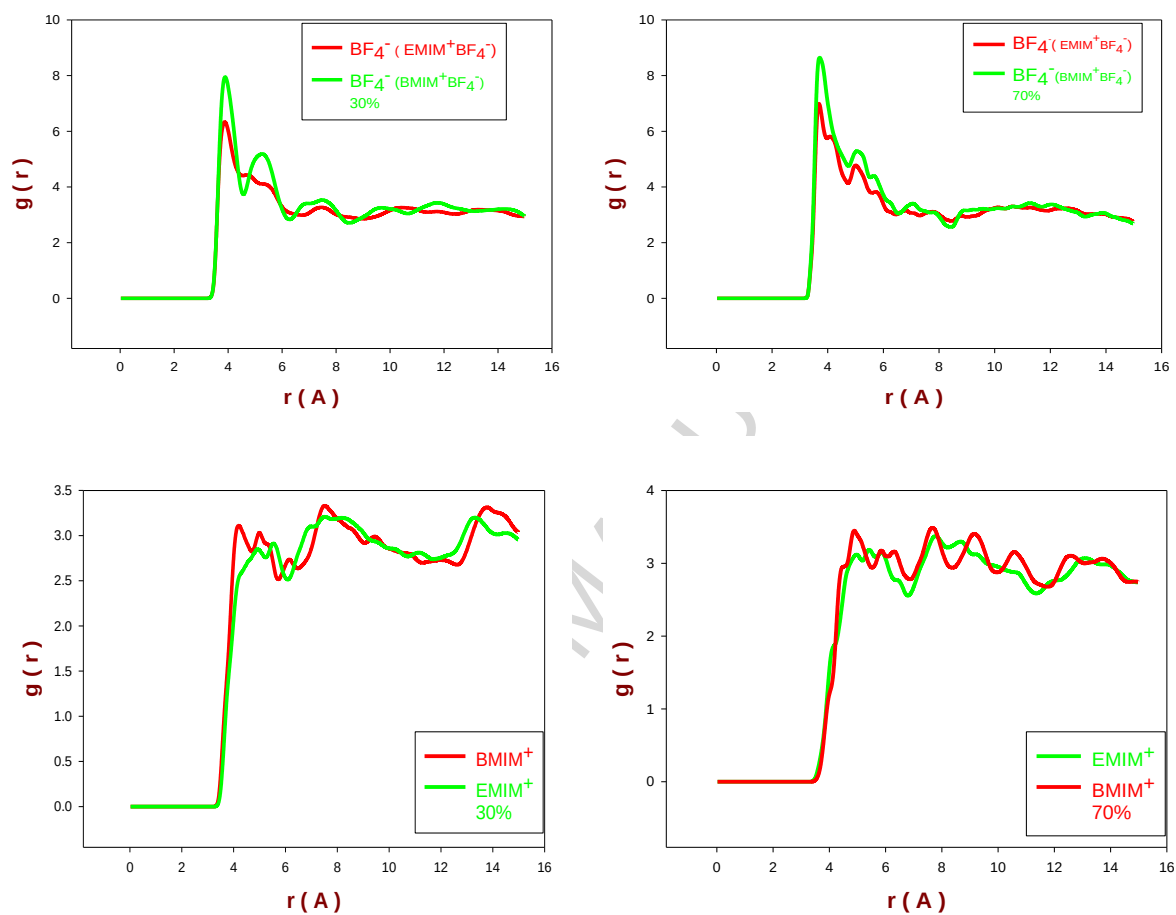


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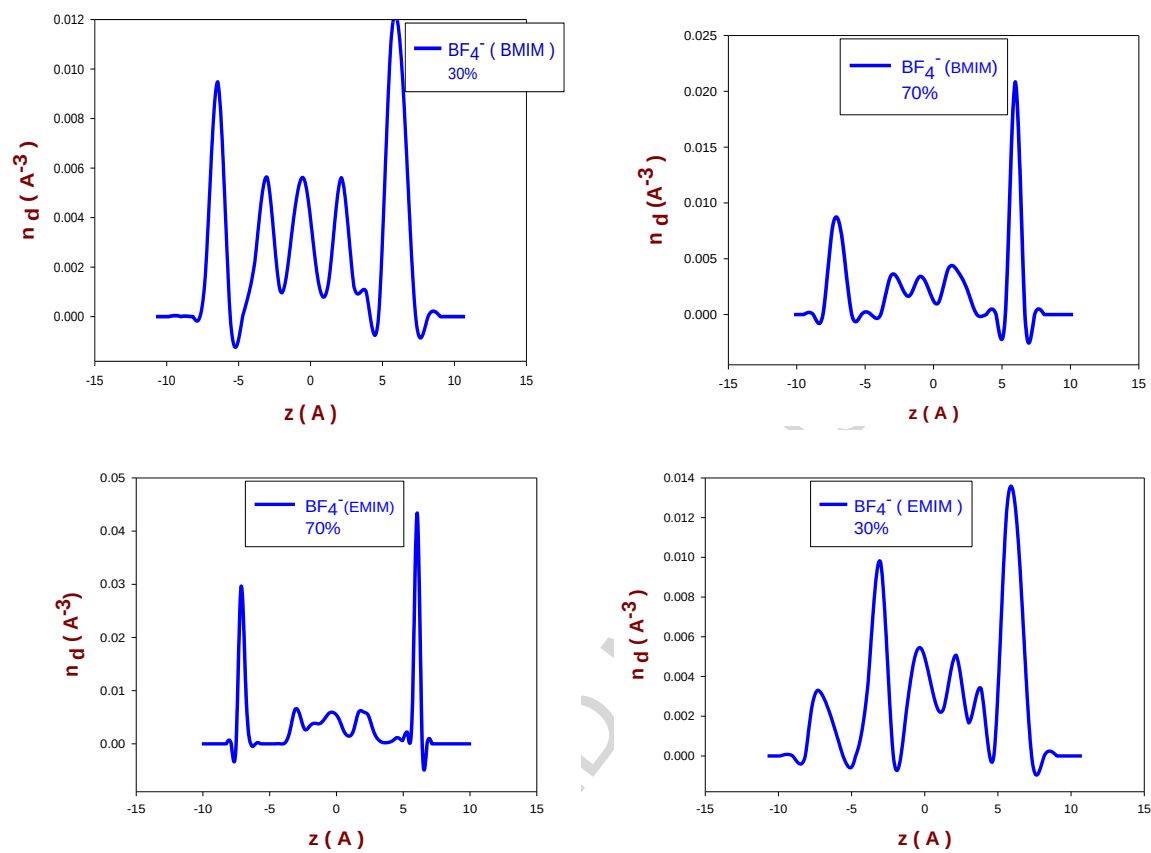


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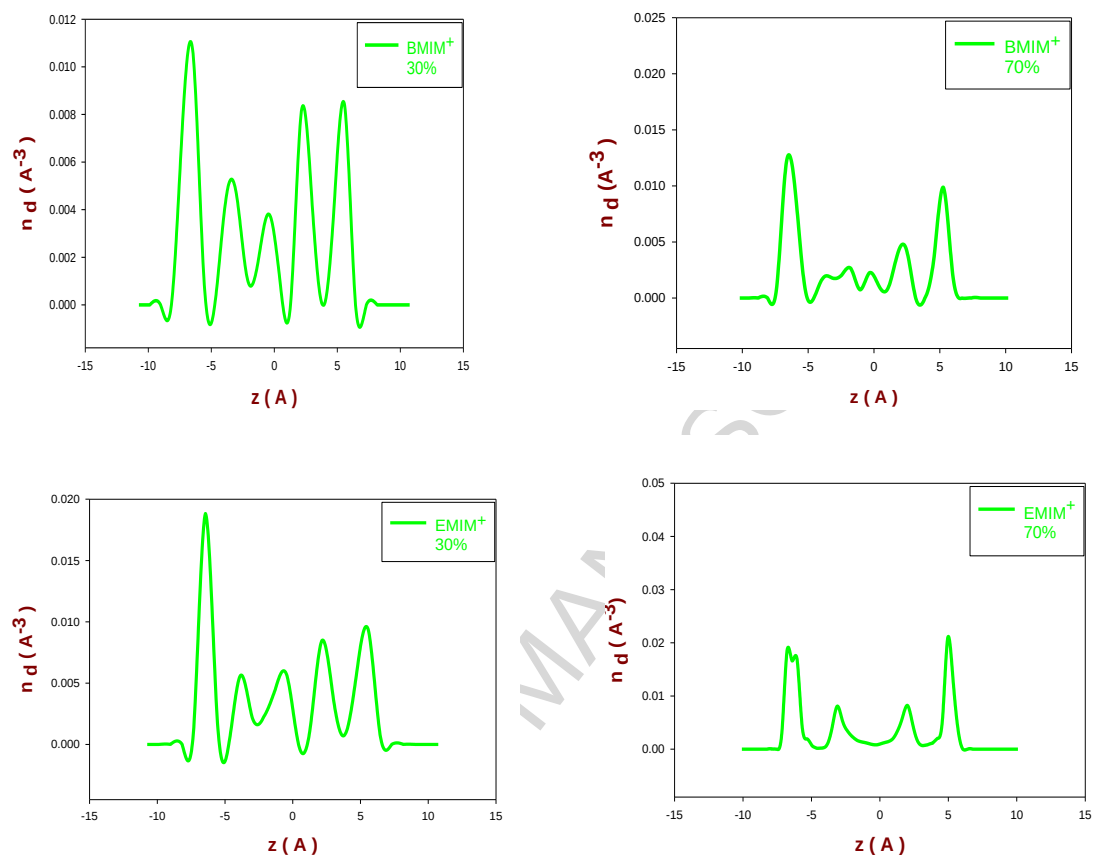


Fig. 15

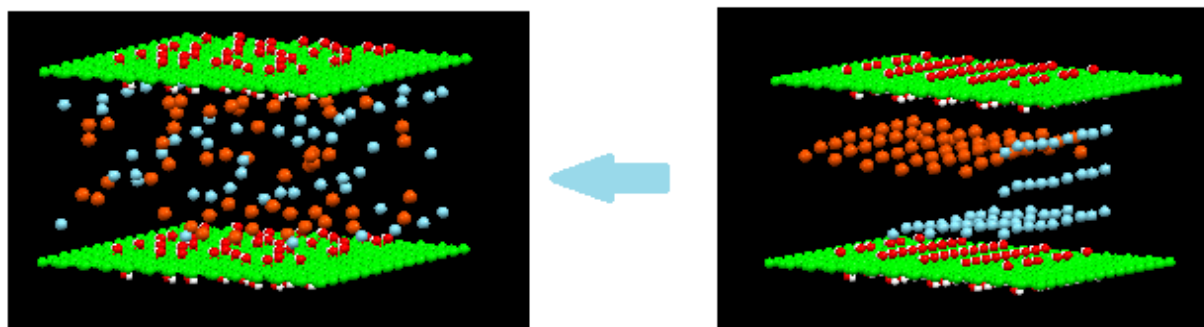


Fig. 16

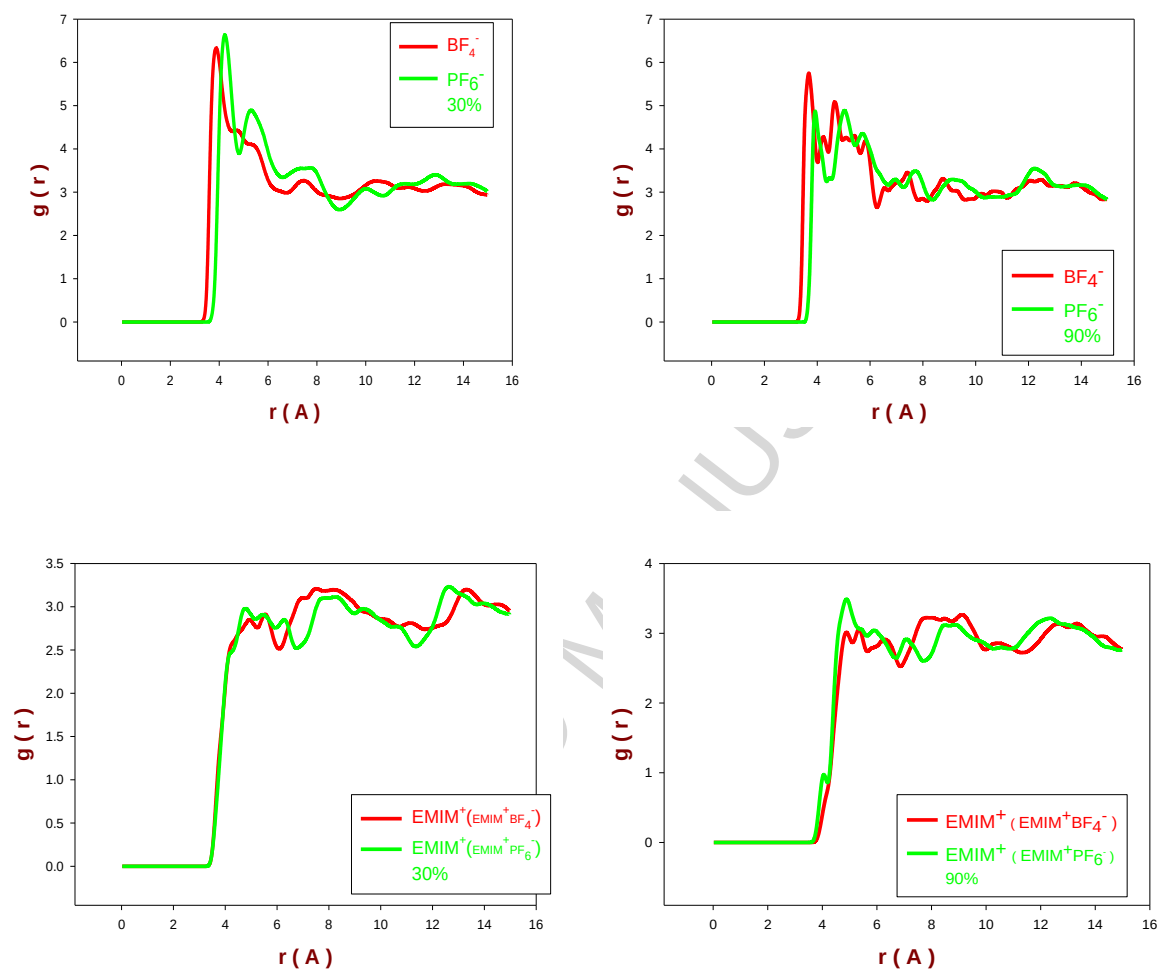


Fig. 17

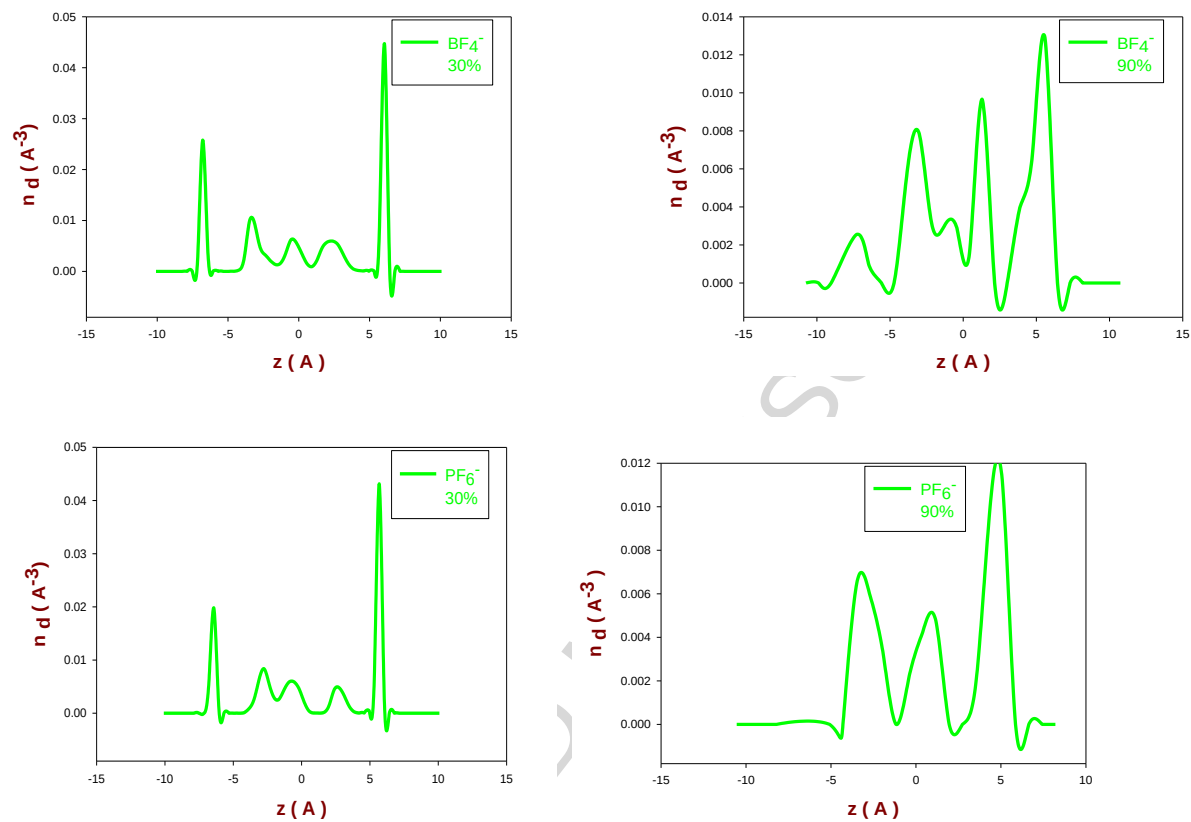
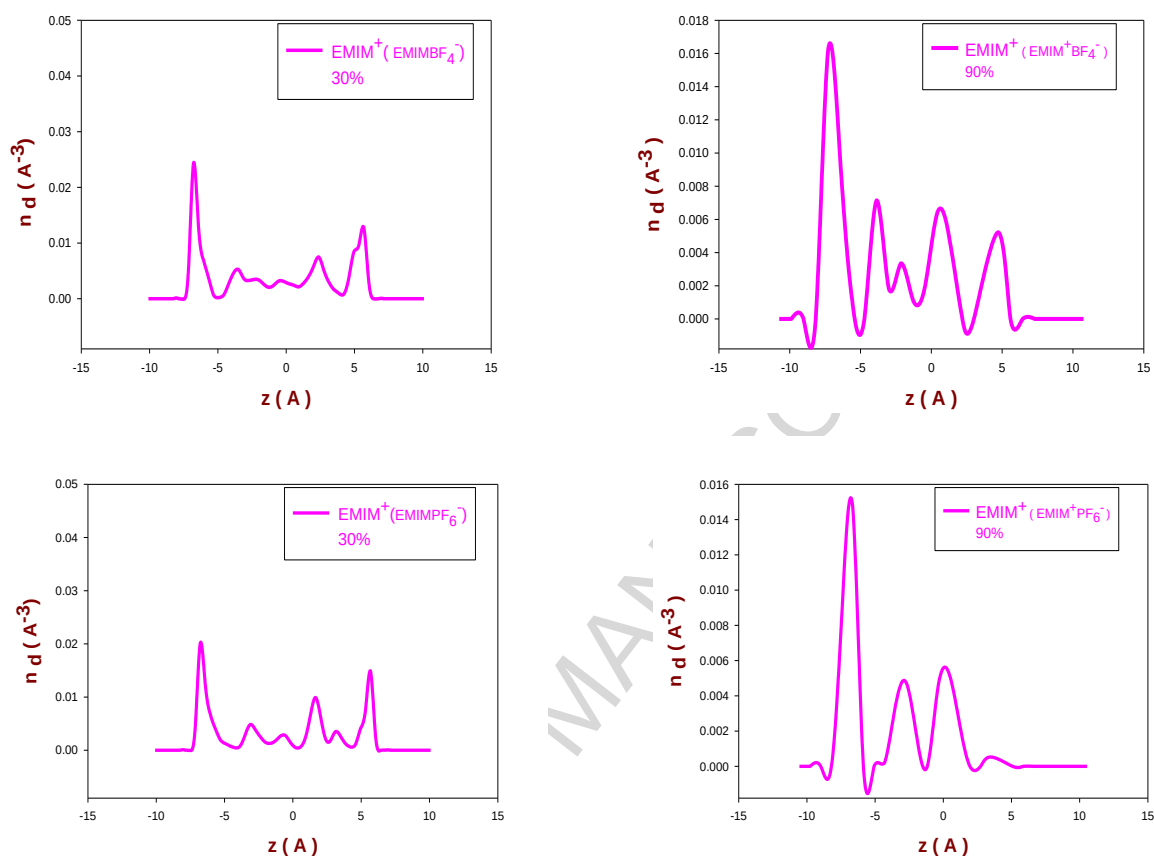


Fig. 18



Highlights

- Absolute value of the configurational energy increases as the percent oxidation of GO increases.
- The "overscreening" phenomenon is the reason of the decreasing of capacity by oxidation degree.
- Energy values for [EMIM][BF₄] are greater than [BMIM][BF₄] at different percent oxidation.